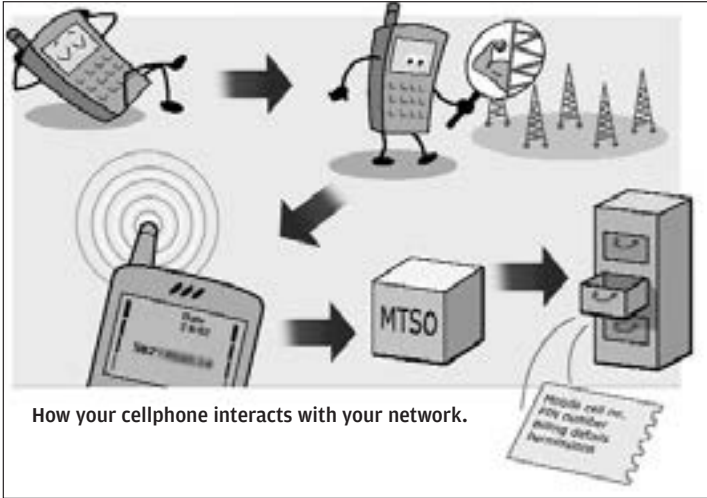




trols all the base stations. Two channels exist in the cellular network. The first is the *Control Channel*, which is used by your MTSO to talk to your phone and vice versa. The second is the *Voice Channel*, on which your conversation is carried.

When you switch on your mobile phone, it scans for control channels within the cell. It's looking for a System Identification Code (SID), which is assigned to each operator and is broadcast by each base station. When the phone receives an SID, it compares it with the one already programmed into it. This SID may be directly programmed in the phone (for a CDMA network) or present on the SIM card (in a GSM network). If the SID it receives is different from the one it has, the phone looks for another control channel. This goes on till it finds an SID that matches. The phone now knows that it has found its 'home' network, and sends it a *Registration Request*, which includes your Mobile Identification Number (MIN) - a unique 10-digit number assigned to your phone by your operator to identify you on the network. Once the MTSO has verified this information, it acknowledges the existence of your phone, and begins to track your position in its database.